

Home Task

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Branch: BE-CSE-AIML

Semester: 5

Subject: ADBMS

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Subject Code: 24CSP-310

Q.1

Julia asked her students to create some coding challenges. Write a query to print the hacker_id, name, and the total number of challenges created by each student. Sort your results by the total number of challenges in descending order. If more than one student created the same number of challenges, then sort the result by hacker_id. If more than one student created the same number of challenges and the count is less than the maximum number of challenges created, then exclude those students from the result.

Input Format

The following tables contain challenge data:

Column	Type
hacker_id	Integer
name	String

- Hackers: The hacker_id is the id of the hacker, and name is the name of the hacker.
- Challenges: The challenge_id is the id of the challenge, and hacker_id is the id of the student who created the challenge.

Column	Type
challenge_id	Integer
hacker_id	Integer

Congratulations

You solved this challenge. Would you like to challenge your friends? [Twitter](#) [LinkedIn](#)

[Next Challenge](#)

Test case 0

Compiler Message

Success

Input (stdin)

1 INPUT

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Expected Output

1 5120 Julia 50
2 18425 Anna 50
3 20023 Brian 50
4 33625 Jason 50
5 41805 Benjamin 50

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Solution:

```
SELECT h.hacker_id, h.name, COUNT(c.challenge_id) AS total_challenges
FROM Hackers h
JOIN Challenges c
  ON h.hacker_id = c.hacker_id
GROUP BY h.hacker_id, h.name
HAVING COUNT(c.challenge_id) = (
  SELECT MAX(challenge_count)
  FROM (
    SELECT COUNT(*) AS challenge_count
    FROM Challenges
    GROUP BY hacker_id
  ) counts
)
OR COUNT(c.challenge_id) IN (
  SELECT challenge_count
  FROM (
    SELECT hacker_id, COUNT(*) AS challenge_count
    FROM Challenges
    GROUP BY hacker_id
  ) counts
  GROUP BY challenge_count
  HAVING COUNT(*) = 1
)
ORDER BY total_challenges DESC, h.hacker_id;
```

Q2

You did such a great job helping Julia with her last coding contest challenge that she wants you to work on this one, too!

The total score of a hacker is the sum of their maximum scores for all of the challenges. Write a query to print the hacker_id, name, and total score of the hackers ordered by the descending score. If more than one hacker achieved the same total score, then sort the result by ascending hacker_id. Exclude all hackers with a total score of 0 from your result.

Input Format

The following tables contain contest data:

Column	Type
hacker_id	Integer
name	String

- Hackers: The hacker_id is the id of the hacker, and name is the name of the hacker.
- Submissions: The submission_id is the id of the submission, hacker_id is the id of the hacker who made the submission, challenge_id is the id of

Column	Type
submission_id	Integer
hacker_id	Integer
challenge_id	Integer
score	Integer

the challenge for which the submission belongs to, and score is the score of the submission.

Congratulations

You solved this challenge. Would you like to challenge your friends? [Twitter](#) [LinkedIn](#)

[Next Challenge](#)

✓ Test case 0

Compiler Message

Success

Input (stdin)

1 INPUT

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Expected Output

1 76971 Ashley 760
2 84200 Susan 710
3 76615 Ryan 700
4 82382 Sara 640
5 79034 Marilyn 580

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```
1  ✓ SELECT h.hacker_id,  
2      |      h.name,  
3      |      SUM(s.max_score) AS total_score  
4  FROM Hackers h  
5  ✓ JOIN (  
6  ✓      SELECT hacker_id,  
7      |      challenge_id,  
8      |      MAX(score) AS max_score  
9      FROM Submissions  
10     GROUP BY hacker_id, challenge_id  
11  ✓ ) s  
12     ON h.hacker_id = s.hacker_id  
13  GROUP BY h.hacker_id, h.name  
14  HAVING SUM(s.max_score) > 0  
15  ✓ ORDER BY total_score DESC,  
16  |      |      h.hacker_id ASC;
```